Symmetry Tables for Crystallographic and Magnetic Point Groups

After R. R. Birss, *Symmetry and Magnetism* (1966)

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1 Content Overview

This document compiles Tables 3, 4a–4f, 6, and 7 from R. R. Birss, *Symmetry and Magnetism* (Elsevier/ North-Holland, 1966), typeset with proper crystallographic notation.

- **Table 3** (p. 4): The 32 classical crystallographic point groups — Hermann–Mauguin (international) symbols, Schoenflies and Shubnikov symbols, symmetry operations, group order, and generating matrices.
- **Table 4a** (p. 5): Maps each of the 32 point groups to tensor-symbol classes (letters A–U) for four tensor types (polar/axial × even/odd rank).
- **Tables 4b–4f** (p. 6): Independent tensor components for each symbol class at ranks 0–4. Table 4b: rank 0 (scalar); Table 4c: rank 1 (vector); Table 4d: rank 2; Table 4e: rank 3; Table 4f: rank 4.
- **Table 6** (p. 10): The 90 magnetic point groups (types I and III) (90 rows: 32 classical + 58 black-and-white; the 32 grey groups are omitted).
- **Table 7** (p. 13): i-tensor and c-tensor symbol classes for the 90 magnetic point groups, using the same A–U scheme as Table 4a.

2 Notation

Point-group symbols

- **Overbars** denote improper rotations (roto-inversions): $\overline{1}$ = inversion, $\overline{2}$ = mirror reflection, $\overline{3}$, $\overline{4}$, $\overline{6}$ = higher-order roto-inversions.
- **Hermann–Mauguin** (international) symbols: e.g. $\overline{4}2m$, $m\overline{3}m$, $6/mmm$.
- **Schoenflies** symbols: e.g. C_{2v}, D_{4h}, O_h.
- **Shubnikov** symbols: use colons (:) for normal subgroups, dots (·) for semidirect products, slashes (/) for quotients. Overbars as above.

Symmetry operations

- $\pm n_z$ = the pair of proper n-fold rotations about z.
- $\pm \overline{n}_z$ = the pair of n-fold roto-inversions about z.
- $\perp$ = perpendicular to the principal axis (e.g. $3(2\perp)$ = three 2-fold axes perpendicular to the 3-fold axis).
- $n(m)$ groups n equivalent operations of type m.
- 2_{xy}, 2_{-xy} = diagonal 2-fold axes (along [110]- and [$1\overline{1}0$]-type directions); similarly $\overline{2}_{xy}$, $\overline{2}_{-xy}$ for the corresponding mirrors.
- A prime ($'$) indicates the operation is combined with time reversal (Tables 6 and 7). In grouped terms $n(m'\perp)$, the prime is on the operation symbol, not the multiplicity count.

Generating matrices

- $\sigma^{(N)}$: named generating matrices (see Birss §5 for definitions).
- $\sigma'^{(N)}$: generating matrix combined with time reversal (Table 6).

Tensor symbol classes and components (Tables 4a–4f, 7)

- Letters A–U label 21 distinct symmetry classes. Table 4a (and Table 7) assigns each point group a letter for each tensor type; this letter is the row key into Tables 4b–4f.
- A dash (—) in Tables 4a or 7 means the tensor vanishes identically.
- In Tables 4b–4f, column headers are tensor index tuples. Permutation shorthands: (3) = 3 unrestricted permutations; (x·3) = 3 permutations keeping a specific index structure; (x:3) = 3 permutations with the first index fixed; (c4) = 4 cyclic permutations; (xy:6) = 6 permutations preserving the relative order of x and y.

– A coordinate symbol (e.g. xx, xyz) means the component survives as an independent parameter. A minus sign (e.g. $-xy$) means the component equals the negative of the named one. A compound entry (e.g. $yyxx$+$xyyx$+$yxyx$) means the component equals that sum. 0 means the component vanishes.
– Parenthesized tensor symbols such as (E_m) in Table 7 denote alternate reference-axis settings (same abstract group, rotated axes).

3 How to Use These Tables

Finding tensor forms for a classical point group

1. Find the point group in **Table 3** to see its symmetry operations and generating matrices.
2. In **Table 4a**, read the symbol class (A–U) for the desired tensor type (polar/axial, even/odd rank).
3. Append the rank to the letter to form the row label: even rank m: $m = 0 \rightarrow$ Table 4b, $m = 2 \rightarrow$ Table 4d, $m = 4 \rightarrow$ Table 4f; odd rank n: $n = 1 \rightarrow$ Table 4c, $n = 3 \rightarrow$ Table 4e.
4. Look up that row in the appropriate table for the independent components.

Example: Point group 422 has axial-even symbol H_m (Table 4a). For the rank-2 axial tensor, look up row H_2 in Table 4d; for rank-4, row H_4 in Table 4f.

Finding tensor forms for a magnetic point group

1. Find the magnetic point group in **Table 6** to see its symmetry operations, classical subgroup, and generating matrices.
2. In **Table 7**, read the i-tensor and c-tensor symbol classes.
3. Use the same rank-mapping as above (Tables 4b–4f) to find the independent components.

The 32 grey (type II) groups are omitted from Tables 6 and 7: for each, the i-tensor equals that of the classical group and every c-tensor vanishes.

4 Cross-References

From	**To**
Table 3 (point groups)	Table 4a (tensor symbol classes), Table 6 (magnetic extension)
Table 4a (symbol classes)	Tables 4b–4f (tensor components), Table 7 (magnetic extension)
Tables 4b–4f (components)	Table 4a or 7 (which row to look up)
Table 6 (magnetic groups)	Table 3 (classical groups, $\sigma^{(N)}$ notation), Table 7 (tensor symbols)
Table 7 (magnetic tensors)	Table 4a (symbol-class scheme), Tables 4b–4f (components), Table 6 (operations)

Table 3: The 32 Crystallographic Point Groups

System	Intl. (abbrev.)	Intl. (full)	Schoenflies	Shubnikov	Symmetry operations	$\lvert G \rvert$	Generating matrices
Triclinic	1	1	C_1	1	1	1	$\sigma^{(0)}$
Triclinic	$\bar{1}$	$\bar{1}$	C_i (S_2)	$\bar{2}$	1, $\bar{1}$	2	$\sigma^{(1)}$
Monoclinic	2	2	C_2	2	1, 2_z	2	$\sigma^{(3)}$
Monoclinic	m	m	C_s (C_{1h})	m	1, $\bar{2}_z$	2	$\sigma^{(5)}$
Monoclinic	$2/m$	$2/m$	C_{2h}	$2{:}m$	1, $\bar{1}$, 2_z, $\bar{2}_z$	4	$\sigma^{(1)}$, $\sigma^{(3)}$
Orthorhombic	222	222	D_2 (V)	$2{:}2$	1, 2_x, 2_y, 2_z	4	$\sigma^{(2)}$, $\sigma^{(3)}$
Orthorhombic	$mm2$	$mm2$	C_{2v}	$2{\cdot}m$	1, 2_z, $\bar{2}_x$, $\bar{2}_y$	4	$\sigma^{(3)}$, $\sigma^{(4)}$
Orthorhombic	mmm	mmm	D_{2h} (V_h)	$m{\cdot}2{:}m$	1, $\bar{1}$, 2_x, 2_y, 2_z, $\bar{2}_x$, $\bar{2}_y$, $\bar{2}_z$	8	$\sigma^{(1)}$, $\sigma^{(2)}$, $\sigma^{(3)}$
Tetragonal	4	4	C_4	4	1, 2_z, $\pm 4_z$	4	$\sigma^{(7)}$
Tetragonal	$\bar{4}$	$\bar{4}$	S_4	$\bar{4}$	1, 2_z, $\pm\bar{4}_z$	4	$\sigma^{(8)}$
Tetragonal	$4/m$	$4/m$	C_{4h}	$4{:}m$	1, $\bar{1}$, 2_z, $\bar{2}_z$, $\pm 4_z$, $\pm\bar{4}_z$	8	$\sigma^{(1)}$, $\sigma^{(7)}$
Tetragonal	422	422	D_4	$4{:}2$	1, 2_x, 2_y, 2_z, 2_{xy}, 2_{-xy}, $\pm 4_z$	8	$\sigma^{(2)}$, $\sigma^{(7)}$
Tetragonal	$4mm$	$4mm$	C_{4v}	$4{\cdot}m$	1, 2_z, $\bar{2}_x$, $\bar{2}_y$, $\bar{2}_{xy}$, $\bar{2}_{-xy}$, $\pm 4_z$	8	$\sigma^{(4)}$, $\sigma^{(7)}$
Tetragonal	$\bar{4}2m$	$\bar{4}2m$	D_{2d} (V_d)	$\bar{4}{\cdot}m$	1, 2_x, 2_y, 2_z, $\bar{2}_{xy}$, $\bar{2}_{-xy}$, $\pm\bar{4}_z$	8	$\sigma^{(2)}$, $\sigma^{(8)}$
Tetragonal	$4/mmm$	$4/mmm$	D_{4h}	$m{\cdot}4{:}m$	1, $\bar{1}$, 2_x, 2_y, 2_z, 2_{xy}, 2_{-xy}, $\bar{2}_x$, $\bar{2}_y$, $\bar{2}_z$, $\bar{2}_{xy}$, $\bar{2}_{-xy}$, $\pm 4_z$, $\pm\bar{4}_z$	16	$\sigma^{(1)}$, $\sigma^{(2)}$, $\sigma^{(7)}$
Trigonal	3	3	C_3	3	1, $\pm 3_z$	3	$\sigma^{(6)}$
Trigonal	$\bar{3}$	$\bar{3}$	C_{3i} (S_6)	$\bar{6}$	1, $\bar{1}$, $\pm 3_z$, $\pm\bar{3}_z$	6	$\sigma^{(1)}$, $\sigma^{(6)}$
Trigonal	32	32	D_3	$3{:}2$	1, $3(2\perp)$, $\pm 3_z$	6	$\sigma^{(2)}$, $\sigma^{(6)}$
Trigonal	$3m$	$3m$	C_{3v}	$3{\cdot}m$	1, $3(\bar{2}\perp)$, $\pm 3_z$	6	$\sigma^{(4)}$, $\sigma^{(6)}$
Trigonal	$\bar{3}m$	$\bar{3}m$	D_{3d}	$\bar{6}{\cdot}m$	1, $\bar{1}$, $3(2\perp)$, $3(\bar{2}\perp)$, $\pm 3_z$, $\pm\bar{3}_z$	12	$\sigma^{(1)}$, $\sigma^{(2)}$, $\sigma^{(6)}$
Hexagonal	6	6	C_6	6	1, 2_z, $\pm 3_z$, $\pm 6_z$	6	$\sigma^{(3)}$, $\sigma^{(6)}$
Hexagonal	$\bar{6}$	$\bar{6}$	C_{3h}	$3{:}m$	1, $\bar{2}_z$, $\pm 3_z$, $\pm\bar{6}_z$	6	$\sigma^{(5)}$, $\sigma^{(6)}$
Hexagonal	$6/m$	$6/m$	C_{6h}	$6{:}m$	1, $\bar{1}$, 2_z, $\bar{2}_z$, $\pm 3_z$, $\pm\bar{3}_z$, $\pm 6_z$, $\pm\bar{6}_z$	12	$\sigma^{(1)}$, $\sigma^{(3)}$, $\sigma^{(6)}$
Hexagonal	622	622	D_6	$6{:}2$	1, $6(2\perp)$, 2_z, $\pm 3_z$, $\pm 6_z$	12	$\sigma^{(2)}$, $\sigma^{(3)}$, $\sigma^{(6)}$
Hexagonal	$6mm$	$6mm$	C_{6v}	$6{\cdot}m$	1, 2_z, $6(\bar{2}\perp)$, $\pm 3_z$, $\pm 6_z$	12	$\sigma^{(3)}$, $\sigma^{(4)}$, $\sigma^{(6)}$
Hexagonal	$\bar{6}m2$	$\bar{6}m2$	D_{3h}	$m{\cdot}3{:}m$	1, $3(2\perp)$, $3(\bar{2}\perp)$, $\bar{2}_z$, $\pm 3_z$, $\pm\bar{6}_z$	12	$\sigma^{(4)}$, $\sigma^{(5)}$, $\sigma^{(6)}$
Hexagonal	$6/mmm$	$6/mmm$	D_{6h}	$m{\cdot}6{:}m$	1, $\bar{1}$, $6(2\perp)$, 2_z, $6(\bar{2}\perp)$, $\bar{2}_z$, $\pm 3_z$, $\pm\bar{3}_z$, $\pm 6_z$, $\pm\bar{6}_z$	24	$\sigma^{(1)}$, $\sigma^{(2)}$, $\sigma^{(3)}$, $\sigma^{(6)}$
Cubic	23	23	T	$3/2$	1, 3(2), $4(\pm 3)$	12	$\sigma^{(3)}$, $\sigma^{(9)}$
Cubic	$m\bar{3}$	$m\bar{3}$	T_h	$\bar{6}/2$	1, $\bar{1}$, 3(2), $3(\bar{2})$, $4(\pm 3)$, $4(\pm\bar{3})$	24	$\sigma^{(1)}$, $\sigma^{(3)}$, $\sigma^{(9)}$
Cubic	432	432	O	$3/4$	1, 9(2), $4(\pm 3)$, $3(\pm 4)$	24	$\sigma^{(7)}$, $\sigma^{(9)}$
Cubic	$\bar{4}3m$	$\bar{4}3m$	T_d	$3/\bar{4}$	1, 3(2), $6(\bar{2})$, $4(\pm 3)$, $3(\pm\bar{4})$	24	$\sigma^{(8)}$, $\sigma^{(9)}$
Cubic	$m\bar{3}m$	$m\bar{3}m$	O_h	$\bar{6}/4$	1, $\bar{1}$, 9(2), $9(\bar{2})$, $4(\pm 3)$, $4(\pm\bar{3})$, $3(\pm 4)$, $3(\pm\bar{4})$	48	$\sigma^{(1)}$, $\sigma^{(7)}$, $\sigma^{(9)}$

Table 4a: Tensor Symbol Classes for the 32 Point Groups

System	Intl. symbol	Reference axes	Polar even m	Axial even m	Polar odd n	Axial odd n
Triclinic	1	any	A_m	A_m	A_n	A_n
Triclinic	$\overline{1}$	any	A_m	—	—	A_n
Monoclinic	2	$2 \parallel z$	B_m	B_m	B_n	B_n
Monoclinic	m	$\overline{2} \parallel z$	B_m	C_m	C_n	B_n
Monoclinic	$2/m$	$2 \parallel z$	B_m	—	—	B_n
Orthorhombic	222	$2 \parallel x$, $2 \parallel y$	D_m	D_m	D_n	D_n
Orthorhombic	$mm2$	$\overline{2} \parallel x$, $\overline{2} \parallel y$	D_m	E_m	E_n	D_n
Orthorhombic	mmm	$2 \parallel x$, $2 \parallel y$	D_m	—	—	D_n
Tetragonal	4	$4 \parallel z$	F_m	F_m	F_n	F_n
Tetragonal	$\overline{4}$	$\overline{4} \parallel z$	F_m	G_m	G_n	F_n
Tetragonal	$4/m$	$4 \parallel z$	F_m	—	—	F_n
Tetragonal	422	$4 \parallel z$, $2 \parallel y$	H_m	H_m	H_n	H_n
Tetragonal	$4mm$	$4 \parallel z$, $\overline{2} \parallel y$	H_m	I_m	I_n	H_n
Tetragonal	$\overline{4}2m$	$\overline{4} \parallel z$, $2 \parallel y$	H_m	J_m	J_n	H_n
Tetragonal	$4/mmm$	$4 \parallel z$, $2 \parallel y$	H_m	—	—	H_n
Trigonal	3	$3 \parallel z$	K_m	K_m	K_n	K_n
Trigonal	$\overline{3}$	$\overline{3} \parallel z$	K_m	—	—	K_n
Trigonal	32	$3 \parallel z$, $2 \parallel y$	L_m	L_m	L_n	L_n
Trigonal	$3m$	$3 \parallel z$, $\overline{2} \parallel y$	L_m	M_m	M_n	L_n
Trigonal	$\overline{3}m$	$\overline{3} \parallel z$, $2 \parallel y$	L_m	—	—	L_n
Hexagonal	6	$6 \parallel z$	N_m	N_m	N_n	N_n
Hexagonal	$\overline{6}$	$\overline{3} \parallel z$	N_m	O_m	O_n	N_n
Hexagonal	$6/m$	$6 \parallel z$	N_m	—	—	N_n
Hexagonal	622	$6 \parallel z$, $2 \parallel y$	P_m	P_m	P_n	P_n
Hexagonal	$6mm$	$6 \parallel z$, $\overline{2} \parallel y$	P_m	Q_m	Q_n	P_n
Hexagonal	$\overline{6}m2$	$3 \parallel z$, $\overline{2} \parallel y$	P_m	R_m	R_n	P_n
Hexagonal	$6/mmm$	$6 \parallel z$, $2 \parallel y$	P_m	—	—	P_n
Cubic	23	$2 \parallel x$, $2 \parallel y$	S_m	S_m	S_n	S_n
Cubic	$m\overline{3}$	$2 \parallel x$, $2 \parallel y$	S_m	—	—	S_n
Cubic	432	$4 \parallel x$, $4 \parallel y$	T_m	T_m	T_n	T_n
Cubic	$\overline{4}3m$	$\overline{4} \parallel x$, $\overline{4} \parallel y$	T_m	U_m	U_n	T_n
Cubic	$m\overline{3}m$	$4 \parallel x$, $4 \parallel y$	T_m	—	—	T_n

Tables 4b, 4c, 4d: Tensor Components — Ranks 0, 1, 2

Table 4b

Rank 0

$m=0$	x
A_0	x
B_0	x
C_0	0
D_0	x
E_0	0
F_0	x
G_0	0
H_0	x
I_0	0
J_0	0
K_0	x
L_0	x
M_0	0
N_0	x
O_0	0
P_0	x
Q_0	0
R_0	0
S_0	x
T_0	x
U_0	0

Table 4c

Rank 1

$n=1$	x	y	z
A_1	x	y	z
B_1	0	0	z
C_1	x	y	0
D_1	0	0	0
E_1	0	0	z
F_1	0	0	z
G_1	0	0	0
H_1	0	0	0
I_1	0	0	z
J_1	0	0	0
K_1	0	0	z
L_1	0	0	0
M_1	0	0	z
N_1	0	0	z
O_1	0	0	0
P_1	0	0	0
Q_1	0	0	z
R_1	0	0	0
S_1	0	0	0
T_1	0	0	0
U_1	0	0	0

Table 4d

Rank 2

$m=2$	xx	yy	zz	xy	yx	$xz\,(2)$	$yz\,(2)$
A_2	xx	yy	zz	xy	yx	xz	yz
B_2	xx	yy	zz	xy	yx	0	0
C_2	0	0	0	0	0	xz	yz
D_2	xx	yy	zz	0	0	0	0
E_2	0	0	0	xy	yx	0	0
F_2	xx	xx	zz	xy	$-xy$	0	0
G_2	xx	$-xx$	0	xy	xy	0	0
H_2	xx	xx	zz	0	0	0	0
I_2	0	0	0	xy	$-xy$	0	0
J_2	xx	$-xx$	0	0	0	0	0
K_2	xx	xx	zz	xy	$-xy$	0	0
L_2	xx	xx	zz	0	0	0	0
M_2	0	0	0	xy	$-xy$	0	0
N_2	xx	xx	zz	xy	$-xy$	0	0
O_2	0	0	0	0	0	0	0
P_2	xx	xx	zz	0	0	0	0
Q_2	0	0	0	xy	$-xy$	0	0
R_2	0	0	0	0	0	0	0
S_2	xx	xx	xx	0	0	0	0
T_2	xx	xx	xx	0	0	0	0
U_2	0	0	0	0	0	0	0

Table 4e: Tensor Components — Rank 3

Row	xxx	yyy	zzz	xxy (3)	yyx (3)	xxz (3)	yyz (3)	zzx (3)	zzy (3)	xyz	xzy	zxy	yxz	yzx	zyx
A_3	xxx	yyy	zzz	xxy	yyx	xxz	yyz	zzx	zzy	xyz	xzy	zxy	yxz	yzx	zyx
B_3	0	0	zzz	0	0	xxz	yyz	0	0	xyz	xzy	zxy	yxz	yzx	zyx
C_3	xxx	yyy	0	xxy	yyx	0	0	zzx	zzy	0	0	0	0	0	0
D_3	0	0	0	0	0	0	0	0	0	xyz	xzy	zxy	yxz	yzx	zyx
E_3	0	0	zzz	0	0	xxz	yyz	0	0	0	0	0	0	0	0
F_3	0	0	zzz	0	0	xxz	xxz	0	0	xyz	xzy	zxy	$-xyz$	$-xzy$	$-zxy$
G_3	0	0	0	0	0	xxz	$-xxz$	0	0	xyz	xzy	zxy	xyz	xzy	zxy
H_3	0	0	0	0	0	0	0	0	0	xyz	xzy	zxy	$-xyz$	$-xzy$	$-zxy$
I_3	0	0	zzz	0	0	xxz	xxz	0	0	0	0	0	0	0	0
J_3	0	0	0	0	0	0	0	0	0	xyz	xzy	zxy	xyz	xzy	zxy
K_3	xxx	yyy	zzz	$-yyy$	$-xxx$	xxz	xxz	0	0	xyz	xzy	zxy	$-xyz$	$-xzy$	$-zxy$
L_3	0	yyy	0	$-yyy$	0	0	0	0	0	xyz	xzy	zxy	$-xyz$	$-xzy$	$-zxy$
M_3	xxx	0	zzz	0	$-xxx$	xxz	xxz	0	0	0	0	0	0	0	0
N_3	0	0	zzz	0	0	xxz	xxz	0	0	xyz	xzy	zxy	$-xyz$	$-xzy$	$-zxy$
O_3	xxx	yyy	0	$-yyy$	$-xxx$	0	0	0	0	0	0	0	0	0	0
P_3	0	0	0	0	0	0	0	0	0	xyz	xzy	zxy	$-xyz$	$-xzy$	$-zxy$
Q_3	0	0	zzz	0	0	xxz	xxz	0	0	0	0	0	0	0	0
R_3	xxx	0	0	0	$-xxx$	0	0	0	0	0	0	0	0	0	0
S_3	0	0	0	0	0	0	0	0	0	xyz	xzy	xyz	xzy	xyz	xzy
T_3	0	0	0	0	0	0	0	0	0	xyz	$-xyz$	xyz	$-xyz$	xyz	$-xyz$
U_3	0	0	0	0	0	0	0	0	0	xyz	xyz	xyz	xyz	xyz	xyz

Table 4f, Part I: Tensor Components — Rank 4 (columns 1–11)

Row	$xxxx$	$yyyy$	$zzzz$	$xxxy$	$yxxx$ $(x{\cdot}3)$	$yyyx$	$xyyy$ $(y{\cdot}3)$	$xxxz$ (4)	$yyyz$ (4)	$zzzx$ (4)	$zzzy$ (4)
A_4	$xxxx$	$yyyy$	$zzzz$	$xxxy$	$yxxx$	$yyyx$	$xyyy$	$xxxz$	$yyyz$	$zzzx$	$zzzy$
B_4	$xxxx$	$yyyy$	$zzzz$	$xxxy$	$yxxx$	$yyyx$	$xyyy$	0	0	0	0
C_4	0	0	0	0	0	0	0	$xxxz$	$yyyz$	$zzzx$	$zzzy$
D_4	$xxxx$	$yyyy$	$zzzz$	0	0	0	0	0	0	0	0
E_4	0	0	0	$xxxy$	$yxxx$	$yyyx$	$xyyy$	0	0	0	0
F_4	$xxxx$	$xxxx$	$zzzz$	$xxxy$	$yxxx$	$-xxxy$	$-yxxx$	0	0	0	0
G_4	$xxxx$	$-xxxx$	0	$xxxy$	$yxxx$	$xxxy$	$yxxx$	0	0	0	0
H_4	$xxxx$	$xxxx$	$zzzz$	0	0	0	0	0	0	0	0
I_4	0	0	0	$xxxy$	$yxxx$	$-xxxy$	$-yxxx$	0	0	0	0
J_4	$xxxx$	$-xxxx$	0	0	0	0	0	0	0	0	0
K_4	$yyxx{+}xyyx{+}yxyx$	$xxxx$	$zzzz$	$yyxy{+}xyyy{+}yxyy$	$yxxx$	$-xxxy$	$-yxxx$	$xxxz$	$yyyz$	0	0
L_4	$yyxx{+}xyyx{+}yxyx$	$xxxx$	$zzzz$	0	0	0	0	$xxxz$	0	0	0
M_4	0	0	0	$yyxy{+}xyyy{+}yxyy$	$yxxx$	$-xxxy$	$-yxxx$	0	$yyyz$	0	0
N_4	$yyxx{+}xyyx{+}yxyx$	$xxxx$	$zzzz$	$yyxy{+}xyyy{+}yxyy$	$yxxx$	$-xxxy$	$-yxxx$	0	0	0	0
O_4	0	0	0	0	0	0	0	$xxxz$	$yyyz$	0	0
P_4	$yyxx{+}xyyx{+}yxyx$	$xxxx$	$zzzz$	0	0	0	0	0	0	0	0
Q_4	0	0	0	$yyxy{+}xyyy{+}yxyy$	$yxxx$	$-xxxy$	$-yxxx$	0	0	0	0
R_4	0	0	0	0	0	0	0	0	$yyyz$	0	0
S_4	$xxxx$	$xxxx$	$xxxx$	0	0	0	0	0	0	0	0
T_4	$xxxx$	$xxxx$	$xxxx$	0	0	0	0	0	0	0	0
U_4	0	0	0	0	0	0	0	0	0	0	0

Table 4f, Part II: Tensor Components — Rank 4 (columns 12–25)

Row	$xxyy$ (x:3)	$yyxx$ (y:3)	$xxzz$ (x:3)	$zzxx$ (z:3)	$yyzz$ (y:3)	$zzyy$ (z:3)	$xxyz$ ($c4$)	$yxzz$ ($c4$)	$yxxx$ ($c4$)	$yyxz$ ($c4$)	$yxyz$ ($c4$)	$xyyz$ ($c4$)	$zzxy$ (xy:6)	$zzyx$ (yx:6)
A_4	$xxyy$	$yyxx$	$xxzz$	$zzxx$	$yyzz$	$zzyy$	$xxyz$	$yxzz$	$yxxx$	$yyxz$	$yxyz$	$xyyz$	$zzxy$	$zzyx$
B_4	$xxyy$	$yyxx$	$xxzz$	$zzxx$	$yyzz$	$zzyy$	0	0	0	0	0	0	$zzxy$	$zzyx$
C_4	0	0	0	0	0	0	$xxyz$	$yxzz$	$yxxx$	$yyxz$	$yxyz$	$xyyz$	0	0
D_4	$xxyy$	$yyxx$	$xxzz$	$zzxx$	$yyzz$	$zzyy$	0	0	0	0	0	0	0	0
E_4	0	0	0	0	0	0	0	0	0	0	0	0	$zzxy$	$zzyx$
F_4	$xxyy$	$xxyy$	$xxzz$	$zzxx$	$xxzz$	$zzxx$	0	0	0	0	0	0	$zzxy$	$-zzxy$
G_4	$xxyy$	$-xxyy$	$xxzz$	$zzxx$	$-xxzz$	$-zzxx$	0	0	0	0	0	0	$zzxy$	$zzxy$
H_4	$xxyy$	$xxyy$	$xxzz$	$zzxx$	$xxzz$	$zzxx$	0	0	0	0	0	0	0	0
I_4	0	0	0	0	0	0	0	0	0	0	0	0	$zzxy$	$-zzxy$
J_4	$xxyy$	$-xxyy$	$xxzz$	$zzxx$	$-xxzz$	$-zzxx$	0	0	0	0	0	0	0	0
K_4	$xxyy$	$xxyy$	$xxzz$	$zzxx$	$xxzz$	$zzxx$	$-yyyy$	$-yyyy$	$-yyyy$	$-xxxx$	$-xxxx$	$-xxxx$	$zzxy$	$-zzxy$
L_4	$xxyy$	$xxyy$	$xxzz$	$zzxx$	$xxzz$	$zzxx$	0	0	0	$-xxxx$	$-xxxx$	$-xxxx$	0	0
M_4	0	0	0	0	0	0	$-yyyy$	$-yyyy$	$-yyyy$	0	0	0	$zzxy$	$-zzxy$
N_4	$xxyy$	$xxyy$	$xxzz$	$zzxx$	$xxzz$	$zzxx$	0	0	0	0	0	0	$zzxy$	$-zzxy$
O_4	0	0	0	0	0	0	$-yyyy$	$-yyyy$	$-yyyy$	$-xxxx$	$-xxxx$	$-xxxx$	0	0
P_4	$xxyy$	$xxyy$	$xxzz$	$zzxx$	$xxzz$	$zzxx$	0	0	0	0	0	0	0	0
Q_4	0	0	0	0	0	0	0	0	0	0	0	0	$zzxy$	$-zzxy$
R_4	0	0	0	0	0	0	$-yyyy$	$-yyyy$	$-yyyy$	0	0	0	0	0
S_4	$xxyy$	$yyxx$	$yyxx$	$xxyy$	$xxyy$	$yyxx$	0	0	0	0	0	0	0	0
T_4	$xxyy$	$xxyy$	$xxyy$	$xxyy$	$xxyy$	$xxyy$	0	0	0	0	0	0	0	0
U_4	$xxyy$	$-xxyy$	$-xxyy$	$xxyy$	$xxyy$	$-xxyy$	0	0	0	0	0	0	0	0

Table 6: The 90 Magnetic Point Groups (Types I and III)

System	Magnetic point group	Shubnikov	Classical subgroup	Shubnikov (subgr.)	Symmetry operators	Gen. matrices (subgroup)	Additional gen. matrix
Triclinic	1	1	—	—	1	$\sigma^{(0)}$	—
Triclinic	$\bar{1}$	$\bar{2}$	—	—	$1,\ \bar{1}$	$\sigma^{(1)}$	—
Triclinic	$\bar{1}'$	$\bar{2}'$	1	1	$1,\ \bar{1}'$	$\sigma^{(0)}$	$\sigma'^{(1)}$
Monoclinic	2	2	—	—	$1,\ 2_z$	$\sigma^{(3)}$	—
Monoclinic	$2'$	$2'$	1	1	$1,\ 2'_z$	$\sigma^{(0)}$	$\sigma'^{(3)}$
Monoclinic	m	m	—	—	$1,\ \bar{2}_z$	$\sigma^{(5)}$	—
Monoclinic	m'	m'	1	1	$1,\ \bar{2}'_z$	$\sigma^{(0)}$	$\sigma'^{(5)}$
Monoclinic	$2/m$	$2{:}m$	—	—	$1,\ \bar{1},\ 2_z,\ \bar{2}_z$	$\sigma^{(1)},\ \sigma^{(3)}$	—
Monoclinic	$2'/m'$	$2'{:}m'$	$\bar{1}$	$\bar{2}$	$1,\ \bar{1},\ 2'_z,\ \bar{2}'_z$	$\sigma^{(1)}$	$\sigma'^{(3)}$
Monoclinic	$2/m'$	$2{:}m'$	2	2	$1,\ 2_z,\ \bar{1}',\ \bar{2}'_z$	$\sigma^{(3)}$	$\sigma'^{(1)}$
Monoclinic	$2'/m$	$2'{:}m$	m	m	$1,\ \bar{2}_z,\ \bar{1}',\ 2'_z$	$\sigma^{(5)}$	$\sigma'^{(1)}$
Orthorhombic	222	$2{:}2$	—	—	$1,\ 2_x,\ 2_y,\ 2_z$	$\sigma^{(2)},\ \sigma^{(3)}$	—
Orthorhombic	$2'2'2$	$2{:}2'$	2	2	$1,\ 2_z,\ 2'_x,\ 2'_y$	$\sigma^{(3)}$	$\sigma'^{(2)}$
Orthorhombic	$mm2$	$2{\cdot}m$	—	—	$1,\ 2_z,\ \bar{2}_x,\ \bar{2}_y$	$\sigma^{(3)},\ \sigma^{(4)}$	—
Orthorhombic	$m'm'2$	$2{\cdot}m'$	2	2	$1,\ 2_z,\ \bar{2}'_x,\ \bar{2}'_y$	$\sigma^{(3)}$	$\sigma'^{(4)}$
Orthorhombic	$2'm'm$	$2'{\cdot}m$	m	m	$1,\ \bar{2}_z,\ 2'_x,\ \bar{2}'_y$	$\sigma^{(5)}$	$\sigma'^{(4)}$
Orthorhombic	mmm	$m{\cdot}2{:}m$	—	—	$1,\ \bar{1},\ 2_x,\ 2_y,\ 2_z,\ \bar{2}_x,\ \bar{2}_y,\ \bar{2}_z$	$\sigma^{(1)},\ \sigma^{(2)},\ \sigma^{(3)}$	—
Orthorhombic	$m'm'm$	$m'{\cdot}2{:}m$	$2/m$	$2{:}m$	$1,\ \bar{1},\ 2_z,\ \bar{2}_z,\ 2'_x,\ 2'_y,\ \bar{2}'_x,\ \bar{2}'_y$	$\sigma^{(1)},\ \sigma^{(3)}$	$\sigma'^{(2)}$
Orthorhombic	$m'm'm'$	$m'{\cdot}2{:}m'$	222	$2{:}2$	$1,\ 2_x,\ 2_y,\ 2_z,\ \bar{1}',\ \bar{2}'_x,\ \bar{2}'_y,\ \bar{2}'_z$	$\sigma^{(2)},\ \sigma^{(3)}$	$\sigma'^{(1)}$
Orthorhombic	mmm'	$m{\cdot}2{:}m'$	$mm2$	$2{\cdot}m$	$1,\ 2_z,\ \bar{2}_x,\ \bar{2}_y,\ \bar{1}',\ 2'_x,\ 2'_y,\ \bar{2}'_z$	$\sigma^{(3)},\ \sigma^{(4)}$	$\sigma'^{(1)}$
Tetragonal	4	4	—	—	$1,\ 2_z,\ \pm 4_z$	$\sigma^{(7)}$	—
Tetragonal	$4'$	$4'$	2	2	$1,\ 2_z,\ \pm 4'_z$	$\sigma^{(3)}$	$\sigma'^{(7)}$
Tetragonal	$\bar{4}$	$\bar{4}$	—	—	$1,\ 2_z,\ \pm\bar{4}_z$	$\sigma^{(8)}$	—
Tetragonal	$\bar{4}'$	$\bar{4}'$	2	2	$1,\ 2_z,\ \pm\bar{4}'_z$	$\sigma^{(3)}$	$\sigma'^{(8)}$
Tetragonal	$4/m$	$4{:}m$	—	—	$1,\ \bar{1},\ 2_z,\ \bar{2}_z,\ \pm 4_z,\ \pm\bar{4}_z$	$\sigma^{(1)},\ \sigma^{(7)}$	—
Tetragonal	$4'/m$	$4'{:}m$	$2/m$	$2{:}m$	$1,\ \bar{1},\ 2_z,\ \bar{2}_z,\ \pm 4'_z,\ \pm\bar{4}'_z$	$\sigma^{(1)},\ \sigma^{(3)}$	$\sigma'^{(7)}$
Tetragonal	$4/m'$	$4{:}m'$	4	4	$1,\ 2_z,\ \pm 4_z,\ \bar{1}',\ \bar{2}'_z,\ \pm\bar{4}'_z$	$\sigma^{(7)}$	$\sigma'^{(1)}$
Tetragonal	$4'/m'$	$4'{:}m'$	$\bar{4}$	$\bar{4}$	$1,\ 2_z,\ \pm\bar{4}_z,\ \bar{1}',\ \bar{2}'_z,\ \pm 4'_z$	$\sigma^{(8)}$	$\sigma'^{(1)}$

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System	Magnetic point group	Shubnikov	Classical subgroup	Shubnikov (subgr.)	Symmetry operators	Gen. matrices (subgroup)	Additional gen. matrix
Tetragonal	422	$4{:}2$	—	—	$1,\ 2_x,\ 2_y,\ 2_z,\ 2_{xy},\ 2_{-xy},\ \pm 4_z$	$\sigma^{(2)},\ \sigma^{(7)}$	—
Tetragonal	$4'22'$	$4'{:}2$	222	$2{:}2$	$1,\ 2_x,\ 2_y,\ 2_z,\ 2'_{xy},\ 2'_{-xy},\ \pm 4'_z$	$\sigma^{(2)},\ \sigma^{(3)}$	$\sigma'^{(7)}$
Tetragonal	$42'2'$	$4{:}2'$	4	4	$1,\ 2_z,\ \pm 4_z,\ 2'_x,\ 2'_y,\ 2'_{xy},\ 2'_{-xy}$	$\sigma^{(7)}$	$\sigma'^{(2)}$
Tetragonal	$4mm$	$4{\cdot}m$	—	—	$1,\ 2_z,\ \overline{2}_x,\ \overline{2}_y,\ \overline{2}_{xy},\ \overline{2}_{-xy},\ \pm 4_z$	$\sigma^{(4)},\ \sigma^{(7)}$	—
Tetragonal	$4'mm'$	$4'{\cdot}m$	$mm2$	$2{\cdot}m$	$1,\ 2_z,\ \overline{2}_x,\ \overline{2}_y,\ \overline{2}'_{xy},\ \overline{2}'_{-xy},\ \pm 4'_z$	$\sigma^{(3)},\ \sigma^{(4)}$	$\sigma'^{(7)}$
Tetragonal	$4m'm'$	$4{\cdot}m'$	4	4	$1,\ 2_z,\ \pm 4_z,\ \overline{2}'_x,\ \overline{2}'_y,\ \overline{2}'_{xy},\ \overline{2}'_{-xy}$	$\sigma^{(7)}$	$\sigma'^{(4)}$
Tetragonal	$\overline{4}2m$	$\overline{4}{\cdot}m$	—	—	$1,\ 2_x,\ 2_y,\ 2_z,\ \overline{2}_{xy},\ \overline{2}_{-xy},\ \pm\overline{4}_z$	$\sigma^{(2)},\ \sigma^{(8)}$	—
Tetragonal	$\overline{4}'2m'$	$\overline{4}'{\cdot}m'$	222	$2{:}2$	$1,\ 2_x,\ 2_y,\ 2_z,\ \overline{2}'_{xy},\ \overline{2}'_{-xy},\ \pm\overline{4}'_z$	$\sigma^{(2)},\ \sigma^{(3)}$	$\sigma'^{(8)}$
Tetragonal	$\overline{4}'m2'$	$\overline{4}'{\cdot}m$	$mm2$	$2{\cdot}m$	$1,\ 2_z,\ \overline{2}_x,\ \overline{2}_y,\ 2'_{xy},\ 2'_{-xy},\ \pm\overline{4}'_z$	$\sigma^{(3)},\ \sigma^{(4)}$	$\sigma'^{(8)}$
Tetragonal	$\overline{4}2'm'$	$\overline{4}{\cdot}m'$	$\overline{4}$	$\overline{4}$	$1,\ 2_z,\ \pm\overline{4}_z,\ 2'_x,\ 2'_y,\ \overline{2}'_{xy},\ \overline{2}'_{-xy}$	$\sigma^{(8)}$	$\sigma'^{(2)}$
Tetragonal	$4/mmm$	$m{\cdot}4{:}m$	—	—	$1,\ \overline{1},\ 2_x,\ 2_y,\ 2_z,\ 2_{xy},\ 2_{-xy},\ \overline{2}_x,\ \overline{2}_y,\ \overline{2}_z,\ \overline{2}_{xy},\ \overline{2}_{-xy},\ \pm 4_z,\ \pm\overline{4}_z$	$\sigma^{(1)},\ \sigma^{(2)},\ \sigma^{(7)}$	—
Tetragonal	$4'/mmm'$	$m{\cdot}4'{:}m$	mmm	$m{\cdot}2{:}m$	$1,\ \overline{1},\ 2_x,\ 2_y,\ 2_z,\ \overline{2}_x,\ \overline{2}_y,\ \overline{2}_z,\ 2'_{xy},\ 2'_{-xy},\ \overline{2}'_{xy},\ \overline{2}'_{-xy},\ \pm 4'_z,\ \pm\overline{4}'_z$	$\sigma^{(1)},\ \sigma^{(2)},\ \sigma^{(3)}$	$\sigma'^{(7)}$
Tetragonal	$4/mm'm'$	$m{\cdot}4{:}m'$	$4/m$	$4{:}m$	$1,\ \overline{1},\ 2_z,\ \overline{2}_z,\ \pm 4_z,\ \pm\overline{4}_z,\ 2'_x,\ 2'_y,\ 2'_{xy},\ 2'_{-xy},\ \overline{2}'_x,\ \overline{2}'_y,\ \overline{2}'_{xy},\ \overline{2}'_{-xy}$	$\sigma^{(1)},\ \sigma^{(7)}$	$\sigma'^{(2)}$
Tetragonal	$4/m'm'm'$	$m'{\cdot}4{:}m'$	422	$4{:}2$	$1,\ 2_x,\ 2_y,\ 2_z,\ 2_{xy},\ 2_{-xy},\ \pm 4_z,\ \overline{1}',\ \overline{2}'_x,\ \overline{2}'_y,\ \overline{2}'_z,\ \overline{2}'_{xy},\ \overline{2}'_{-xy},\ \pm\overline{4}'_z$	$\sigma^{(2)},\ \sigma^{(7)}$	$\sigma'^{(1)}$
Tetragonal	$4/m'mm$	$m'{\cdot}4{:}m$	$4mm$	$4{\cdot}m$	$1,\ 2_z,\ \overline{2}_x,\ \overline{2}_y,\ \overline{2}_{xy},\ \overline{2}_{-xy},\ \pm 4_z,\ \overline{1}',\ 2'_x,\ 2'_y,\ \overline{2}'_z,\ 2'_{xy},\ 2'_{-xy},\ \pm\overline{4}'_z$	$\sigma^{(4)},\ \sigma^{(7)}$	$\sigma'^{(1)}$
Tetragonal	$4'/m'm'm$	$m'{\cdot}4'{:}m'$	$\overline{4}2m$	$\overline{4}{\cdot}m$	$1,\ 2_x,\ 2_y,\ 2_z,\ \overline{2}_{xy},\ \overline{2}_{-xy},\ \pm\overline{4}_z,\ \overline{1}',\ \overline{2}'_x,\ \overline{2}'_y,\ \overline{2}'_z,\ 2'_{xy},\ 2'_{-xy},\ \pm 4'_z$	$\sigma^{(2)},\ \sigma^{(8)}$	$\sigma'^{(1)}$
Trigonal	3	3	—	—	$1,\ \pm 3_z$	$\sigma^{(6)}$	—
Trigonal	$\overline{3}$	$\overline{6}$	—	—	$1,\ \overline{1},\ \pm 3_z,\ \pm\overline{3}_z$	$\sigma^{(1)},\ \sigma^{(6)}$	—
Trigonal	$\overline{3}'$	$\overline{6}'$	3	3	$1,\ \pm 3_z,\ \overline{1}',\ \pm\overline{3}'_z$	$\sigma^{(6)}$	$\sigma'^{(1)}$
Trigonal	32	$3{:}2$	—	—	$1,\ 3(2\perp),\ \pm 3_z$	$\sigma^{(2)},\ \sigma^{(6)}$	—
Trigonal	$32'$	$3{:}2'$	3	3	$1,\ \pm 3_z,\ 3(2'\perp)$	$\sigma^{(6)}$	$\sigma'^{(2)}$
Trigonal	$3m$	$3{\cdot}m$	—	—	$1,\ 3(\overline{2}\perp),\ \pm 3_z$	$\sigma^{(4)},\ \sigma^{(6)}$	—
Trigonal	$3m'$	$3{\cdot}m'$	3	3	$1,\ \pm 3_z,\ 3(\overline{2}'\perp)$	$\sigma^{(6)}$	$\sigma'^{(4)}$
Trigonal	$\overline{3}m$	$\overline{6}{\cdot}m$	—	—	$1,\ \overline{1},\ 3(2\perp),\ 3(\overline{2}\perp),\ \pm 3_z,\ \pm\overline{3}_z$	$\sigma^{(1)},\ \sigma^{(2)},\ \sigma^{(6)}$	—
Trigonal	$\overline{3}m'$	$\overline{6}{\cdot}m'$	$\overline{3}$	$\overline{6}$	$1,\ \overline{1},\ \pm 3_z,\ \pm\overline{3}_z,\ 3(2'\perp),\ 3(\overline{2}'\perp)$	$\sigma^{(1)},\ \sigma^{(6)}$	$\sigma'^{(2)}$
Trigonal	$\overline{3}'m'$	$\overline{6}'{\cdot}m'$	32	$3{:}2$	$1,\ 3(2\perp),\ \pm 3_z,\ \overline{1}',\ 3(\overline{2}'\perp),\ \pm\overline{3}'_z$	$\sigma^{(2)},\ \sigma^{(6)}$	$\sigma'^{(1)}$
Trigonal	$\overline{3}'m$	$\overline{6}'{\cdot}m$	$3m$	$3{\cdot}m$	$1,\ \pm 3_z,\ 3(\overline{2}\perp),\ \overline{1}',\ 3(2'\perp),\ \pm\overline{3}'_z$	$\sigma^{(4)},\ \sigma^{(6)}$	$\sigma'^{(1)}$

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System	Magnetic point group	Shubnikov	Classical subgroup	Shubnikov (subgr.)	Symmetry operators	Gen. matrices (subgroup)	Additional gen. matrix
Hexagonal	6	6	—	—	$1,\ 2_z,\ \pm 3_z,\ \pm 6_z$	$\sigma^{(3)},\ \sigma^{(6)}$	—
Hexagonal	$6'$	$6'$	3	3	$1,\ \pm 3_z,\ 2'_z,\ \pm 6'_z$	$\sigma^{(6)}$	$\sigma'^{(3)}$
Hexagonal	$\overline{6}$	$3{:}m$	—	—	$1,\ \overline{2}_z,\ \pm 3_z,\ \pm\overline{6}_z$	$\sigma^{(5)},\ \sigma^{(6)}$	—
Hexagonal	$\overline{6}'$	$3{:}m'$	3	3	$1,\ \pm 3_z,\ \overline{2}'_z,\ \pm\overline{6}'_z$	$\sigma^{(6)}$	$\sigma'^{(5)}$
Hexagonal	$6/m$	$6{:}m$	—	—	$1,\ \overline{1},\ 2_z,\ \overline{2}_z,\ \pm 3_z,\ \pm\overline{3}_z,\ \pm 6_z,\ \pm\overline{6}_z$	$\sigma^{(1)},\ \sigma^{(3)},\ \sigma^{(6)}$	—
Hexagonal	$6'/m'$	$6'{:}m'$	$\overline{3}$	$\overline{6}$	$1,\ \overline{1},\ \pm 3_z,\ \pm\overline{3}_z,\ 2'_z,\ \overline{2}'_z,\ \pm 6'_z,\ \pm\overline{6}'_z$	$\sigma^{(1)},\ \sigma^{(6)}$	$\sigma'^{(3)}$
Hexagonal	$6/m'$	$6{:}m'$	6	6	$1,\ 2_z,\ \pm 3_z,\ \pm 6_z,\ \overline{1}',\ \overline{2}'_z,\ \pm\overline{3}'_z,\ \pm\overline{6}'_z$	$\sigma^{(3)},\ \sigma^{(6)}$	$\sigma'^{(1)}$
Hexagonal	$6'/m$	$6'{:}m$	$\overline{6}$	$3{:}m$	$1,\ \overline{2}_z,\ \pm 3_z,\ \pm\overline{6}_z,\ \overline{1}',\ 2'_z,\ \pm\overline{3}'_z,\ \pm 6'_z$	$\sigma^{(5)},\ \sigma^{(6)}$	$\sigma'^{(1)}$
Hexagonal	622	$6{:}2$	—	—	$1,\ 6(2\perp),\ 2_z,\ \pm 3_z,\ \pm 6_z$	$\sigma^{(2)},\ \sigma^{(3)},\ \sigma^{(6)}$	—
Hexagonal	$6'22'$	$6'{:}2$	32	$3{:}2$	$1,\ 3(2\perp),\ \pm 3_z,\ 3(2'\perp),\ 2'_z,\ \pm 6'_z$	$\sigma^{(2)},\ \sigma^{(6)}$	$\sigma'^{(3)}$
Hexagonal	$62'2'$	$6{:}2'$	6	6	$1,\ 2_z,\ \pm 3_z,\ \pm 6_z,\ 6(2'\perp)$	$\sigma^{(3)},\ \sigma^{(6)}$	$\sigma'^{(2)}$
Hexagonal	$6mm$	$6{\cdot}m$	—	—	$1,\ 2_z,\ 6(\overline{2}\perp),\ \pm 3_z,\ \pm 6_z$	$\sigma^{(3)},\ \sigma^{(4)},\ \sigma^{(6)}$	—
Hexagonal	$6'mm'$	$6'{\cdot}m$	$3m$	$3{\cdot}m$	$1,\ 3(\overline{2}\perp),\ \pm 3_z,\ 2'_z,\ \pm 6'_z,\ 3(\overline{2}'\perp)$	$\sigma^{(4)},\ \sigma^{(6)}$	$\sigma'^{(3)}$
Hexagonal	$6m'm'$	$6{\cdot}m'$	6	6	$1,\ 2_z,\ \pm 3_z,\ \pm 6_z,\ 6(\overline{2}'\perp)$	$\sigma^{(3)},\ \sigma^{(6)}$	$\sigma'^{(4)}$
Hexagonal	$\overline{6}m2$	$m{\cdot}3{:}m$	—	—	$1,\ 3(2\perp),\ 3(\overline{2}\perp),\ \overline{2}_z,\ \pm 3_z,\ \pm\overline{6}_z$	$\sigma^{(4)},\ \sigma^{(5)},\ \sigma^{(6)}$	—
Hexagonal	$\overline{6}'2m'$	$m'{\cdot}3{:}m'$	32	$3{:}2$	$1,\ 3(2\perp),\ \pm 3_z,\ \overline{2}'_z,\ \pm\overline{6}'_z,\ 3(\overline{2}'\perp)$	$\sigma^{(2)},\ \sigma^{(6)}$	$\sigma'^{(5)}$
Hexagonal	$\overline{6}'m2'$	$m{\cdot}3{:}m'$	$3m$	$3{\cdot}m$	$1,\ 3(\overline{2}\perp),\ \pm 3_z,\ 3(2'\perp),\ \overline{2}'_z,\ \pm\overline{6}'_z$	$\sigma^{(4)},\ \sigma^{(6)}$	$\sigma'^{(5)}$
Hexagonal	$\overline{6}m'2'$	$m'{\cdot}3{:}m$	$\overline{6}$	$3{:}m$	$1,\ \overline{2}_z,\ \pm 3_z,\ \pm\overline{6}_z,\ 3(2'\perp),\ 3(\overline{2}'\perp)$	$\sigma^{(5)},\ \sigma^{(6)}$	$\sigma'^{(4)}$
Hexagonal	$6/mmm$	$m{\cdot}6{:}m$	—	—	$1,\ \overline{1},\ 6(2\perp),\ 2_z,\ 6(\overline{2}\perp),\ \overline{2}_z,\ \pm 3_z,\ \pm\overline{3}_z,\ \pm 6_z,\ \pm\overline{6}_z$	$\sigma^{(1)},\ \sigma^{(2)},\ \sigma^{(3)},\ \sigma^{(6)}$	—
Hexagonal	$6'/m'mm'$	$m'{\cdot}6'{:}m'$	$\overline{3}m$	$\overline{6}{\cdot}m$	$1,\ \overline{1},\ 3(2\perp),\ 3(\overline{2}\perp),\ \pm 3_z,\ \pm\overline{3}_z,\ 3(2'\perp),\ 2'_z,\ 3(\overline{2}'\perp),\ \overline{2}'_z,\ \pm 6'_z,\ \pm\overline{6}'_z$	$\sigma^{(1)},\ \sigma^{(2)},\ \sigma^{(6)}$	$\sigma'^{(3)}$
Hexagonal	$6/mm'm'$	$m'{\cdot}6{:}m$	$6/m$	$6{:}m$	$1,\ \overline{1},\ 2_z,\ \overline{2}_z,\ \pm 3_z,\ \pm\overline{3}_z,\ \pm 6_z,\ \pm\overline{6}_z,\ 6(2'\perp),\ 6(\overline{2}'\perp)$	$\sigma^{(1)},\ \sigma^{(3)},\ \sigma^{(6)}$	$\sigma'^{(2)}$
Hexagonal	$6/m'm'm'$	$m'{\cdot}6{:}m'$	622	$6{:}2$	$1,\ 6(2\perp),\ 2_z,\ \pm 3_z,\ \pm 6_z,\ \overline{1}',\ \overline{2}'_z,\ \pm\overline{3}'_z,\ \pm\overline{6}'_z,\ 6(\overline{2}'\perp)$	$\sigma^{(2)},\ \sigma^{(3)},\ \sigma^{(6)}$	$\sigma'^{(1)}$
Hexagonal	$6/m'mm$	$m{\cdot}6{:}m'$	$6mm$	$6{\cdot}m$	$1,\ 2_z,\ 6(\overline{2}\perp),\ \pm 3_z,\ \pm 6_z,\ \overline{1}',\ \overline{2}'_z,\ 6(2'\perp),\ \pm\overline{3}'_z,\ \pm\overline{6}'_z$	$\sigma^{(3)},\ \sigma^{(4)},\ \sigma^{(6)}$	$\sigma'^{(1)}$
Hexagonal	$6'/mm'm$	$m{\cdot}6'{:}m$	$\overline{6}m2$	$m{\cdot}3{:}m$	$1,\ 3(2\perp),\ 3(\overline{2}\perp),\ \overline{2}_z,\ \pm 3_z,\ \pm\overline{6}_z,\ \overline{1}',\ 2'_z,\ 3(2'\perp),\ 3(\overline{2}'\perp),\ \pm\overline{3}'_z,\ \pm 6'_z$	$\sigma^{(4)},\ \sigma^{(5)},\ \sigma^{(6)}$	$\sigma'^{(1)}$
Cubic	23	$3/2$	—	—	$1,\ 3(2),\ 4(\pm 3)$	$\sigma^{(3)},\ \sigma^{(9)}$	—
Cubic	$m\overline{3}$	$\overline{6}/2$	—	—	$1,\ \overline{1},\ 3(2),\ 3(\overline{2}),\ 4(\pm 3),\ 4(\pm\overline{3})$	$\sigma^{(1)},\ \sigma^{(3)},\ \sigma^{(9)}$	—
Cubic	$m'\overline{3}$	$\overline{6}'/2$	23	$3/2$	$1,\ 3(2),\ 4(\pm 3),\ \overline{1}',\ 3(\overline{2}'),\ 4(\pm\overline{3}')$	$\sigma^{(3)},\ \sigma^{(9)}$	$\sigma'^{(1)}$
Cubic	432	$3/4$	—	—	$1,\ 9(2),\ 4(\pm 3),\ 3(\pm 4)$	$\sigma^{(7)},\ \sigma^{(9)}$	—
Cubic	$4'32'$	$3/4'$	23	$3/2$	$1,\ 3(2),\ 4(\pm 3),\ 6(2'),\ 3(\pm 4')$	$\sigma^{(3)},\ \sigma^{(9)}$	$\sigma'^{(7)}$
Cubic	$\overline{4}3m$	$3/\overline{4}$	—	—	$1,\ 3(2),\ 6(\overline{2}),\ 4(\pm 3),\ 3(\pm\overline{4})$	$\sigma^{(8)},\ \sigma^{(9)}$	—
Cubic	$\overline{4}'3m'$	$3/\overline{4}'$	23	$3/2$	$1,\ 3(2),\ 4(\pm 3),\ 6(\overline{2}'),\ 3(\pm\overline{4}')$	$\sigma^{(3)},\ \sigma^{(9)}$	$\sigma'^{(8)}$
Cubic	$m\overline{3}m$	$\overline{6}/4$	—	—	$1,\ \overline{1},\ 9(2),\ 9(\overline{2}),\ 4(\pm 3),\ 4(\pm\overline{3}),\ 3(\pm 4),\ 3(\pm\overline{4})$	$\sigma^{(1)},\ \sigma^{(7)},\ \sigma^{(9)}$	—
Cubic	$m\overline{3}m'$	$\overline{6}/4'$	$m\overline{3}$	$\overline{6}/2$	$1,\ \overline{1},\ 3(2),\ 3(\overline{2}),\ 4(\pm 3),\ 4(\pm\overline{3}),\ 6(2'),\ 6(\overline{2}'),\ 3(\pm 4'),\ 3(\pm\overline{4}')$	$\sigma^{(1)},\ \sigma^{(3)},\ \sigma^{(9)}$	$\sigma'^{(7)}$
Cubic	$m'\overline{3}m'$	$\overline{6}'/4'$	432	$3/4$	$1,\ 9(2),\ 4(\pm 3),\ 3(\pm 4),\ \overline{1}',\ 9(\overline{2}'),\ 4(\pm\overline{3}'),\ 3(\pm\overline{4}')$	$\sigma^{(7)},\ \sigma^{(9)}$	$\sigma'^{(1)}$
Cubic	$m'\overline{3}m$	$\overline{6}'/4$	$\overline{4}3m$	$3/\overline{4}$	$1,\ 3(2),\ 6(\overline{2}),\ 4(\pm 3),\ 3(\pm\overline{4}),\ \overline{1}',\ 3(\overline{2}'),\ 6(2'),\ 4(\pm\overline{3}'),\ 3(\pm 4')$	$\sigma^{(8)},\ \sigma^{(9)}$	$\sigma'^{(1)}$

Table 7: Tensor Symbol Classes for the Magnetic Point Groups

				i-tensor				c-tensor			
System	Magnetic point group	Assoc. group A	Assoc. group B	Polar even m	Axial even m	Polar odd n	Axial odd n	Polar even m	Axial even m	Polar odd n	Axial odd n
Triclinic	1	—	—	A_m	A_m	A_n	A_n	A_m	A_m	A_n	A_n
Triclinic	$\bar{1}$	—	—	A_m	—	—	A_n	A_m	—	—	A_n
Triclinic	$\bar{1}'$	1	$\bar{1}$	A_m	—	—	A_n	—	A_m	A_n	—
Monoclinic	2	—	—	B_m	B_m	B_n	B_n	B_m	B_m	B_n	B_n
Monoclinic	$2'$	m	m	B_m	B_m	B_n	B_n	C_m	C_m	C_n	C_n
Monoclinic	m	—	—	B_m	C_m	C_n	B_n	B_m	C_m	C_n	B_n
Monoclinic	m'	2	m	B_m	C_m	C_n	B_n	C_m	B_m	B_n	C_n
Monoclinic	$2/m$	—	—	B_m	—	—	B_n	B_m	—	—	B_n
Monoclinic	$2'/m'$	$2/m$	m	B_m	—	—	B_n	C_m	—	—	C_n
Monoclinic	$2/m'$	2	$2/m$	B_m	—	—	B_n	—	B_m	B_n	—
Monoclinic	$2'/m$	m	$2/m$	B_m	—	—	B_n	—	C_m	C_n	—
Orthorhombic	222	—	—	D_m	D_m	D_n	D_n	D_m	D_m	D_n	D_n
Orthorhombic	$2'2'2$	$mm2$	$mm2$	D_m	D_m	D_n	D_n	E_m	E_m	E_n	E_n
Orthorhombic	$mm2$	—	—	D_m	E_m	E_n	D_n	D_m	E_m	E_n	D_n
Orthorhombic	$m'm'2$	222	$mm2$	D_m	E_m	E_n	D_n	E_m	D_m	D_n	E_n
Orthorhombic	$(2'm'm)$	$(m2m)$	$mm2$	D_m	(E_m)	(E_n)	D_n	E_m	(E_m)	(E_n)	E_n
Orthorhombic	mmm	—	—	D_m	—	—	D_n	D_m	—	—	D_n
Orthorhombic	$m'm'm$	mmm	$mm2$	D_m	—	—	D_n	E_m	—	—	E_n
Orthorhombic	$m'm'm'$	222	mmm	D_m	—	—	D_n	—	D_m	D_n	—
Orthorhombic	mmm'	$mm2$	mmm	D_m	—	—	D_n	—	E_m	E_n	—
Tetragonal	4	—	—	F_m	F_m	F_n	F_n	F_m	F_m	F_n	F_n
Tetragonal	$4'$	$\bar{4}$	$\bar{4}$	F_m	F_m	F_n	F_n	G_m	G_m	G_n	G_n
Tetragonal	$\bar{4}$	—	—	F_m	G_m	G_n	F_n	F_m	G_m	G_n	F_n
Tetragonal	$\bar{4}'$	4	$\bar{4}$	F_m	G_m	G_n	F_n	G_m	F_m	F_n	G_n
Tetragonal	$4/m$	—	—	F_m	—	—	F_n	F_m	—	—	F_n
Tetragonal	$4'/m$	$4/m$	$\bar{4}$	F_m	—	—	F_n	G_m	—	—	G_n
Tetragonal	$4/m'$	4	$4/m$	F_m	—	—	F_n	—	F_m	F_n	—
Tetragonal	$4'/m'$	$\bar{4}$	$4/m$	F_m	—	—	F_n	—	G_m	G_n	—

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System	Magnetic point group	Assoc. group A	Assoc. group B	**i-tensor** Polar even m	**i-tensor** Axial even m	**i-tensor** Polar odd n	**i-tensor** Axial odd n	**c-tensor** Polar even m	**c-tensor** Axial even m	**c-tensor** Polar odd n	**c-tensor** Axial odd n
Tetragonal	422	—	—	H_m	H_m	H_n	H_n	H_m	H_m	H_n	H_n
Tetragonal	$4'22'$	$\overline{4}2m$	$\overline{4}2m$	H_m	H_m	H_n	H_n	J_m	J_m	J_n	J_n
Tetragonal	$42'2'$	$4mm$	$4mm$	H_m	H_m	H_n	H_n	I_m	I_m	I_n	I_n
Tetragonal	$4mm$	—	—	H_m	I_m	I_n	H_n	H_m	I_m	I_n	H_n
Tetragonal	$4'mm'$	$(\overline{4}m2)$	$\overline{4}2m$	H_m	I_m	I_n	H_n	J_m	(J_m)	(J_n)	J_n
Tetragonal	$4m'm'$	422	$4mm$	H_m	I_m	I_n	H_n	I_m	H_m	H_n	I_n
Tetragonal	$\overline{4}2m$	—	—	H_m	J_m	J_n	H_n	H_m	J_m	J_n	H_n
Tetragonal	$\overline{4}'2m'$	422	$\overline{4}2m$	H_m	J_m	J_n	H_n	J_m	H_m	H_n	J_n
Tetragonal	$(\overline{4}'m2')$	$4mm$	$\overline{4}2m$	H_m	(J_m)	(J_n)	H_n	J_m	I_m	I_n	J_n
Tetragonal	$\overline{4}2'm'$	$(\overline{4}2m)$	$4mm$	H_m	J_m	J_n	H_n	I_m	(J_m)	(J_n)	I_n
Tetragonal	$4/mmm$	—	—	H_m	—	—	H_n	H_m	—	—	H_n
Tetragonal	$4'/mmm'$	$4/mmm$	$\overline{4}2m$	H_m	—	—	H_n	J_m	—	—	J_n
Tetragonal	$4/mm'm'$	$4/mmm$	$4mm$	H_m	—	—	H_n	I_m	—	—	I_n
Tetragonal	$4/m'm'm'$	422	$4/mmm$	H_m	—	—	H_n	—	H_m	H_n	—
Tetragonal	$4/m'mm$	$4mm$	$4/mmm$	H_m	—	—	H_n	—	I_m	I_n	—
Tetragonal	$4'/m'm'm$	$\overline{4}2m$	$4/mmm$	H_m	—	—	H_n	—	J_m	J_n	—
Trigonal	3	—	—	K_m	K_m	K_n	K_n	K_m	K_m	K_n	K_n
Trigonal	$\overline{3}$	—	—	K_m	—	—	K_n	K_m	—	—	K_n
Trigonal	$\overline{3}'$	3	$\overline{3}$	K_m	—	—	K_n	—	K_m	K_n	—
Trigonal	32	—	—	L_m	L_m	L_n	L_n	L_m	L_m	L_n	L_n
Trigonal	$32'$	$3m$	$3m$	L_m	L_m	L_n	L_n	M_m	M_m	M_n	M_n
Trigonal	$3m$	—	—	L_m	M_m	M_n	L_n	L_m	M_m	M_n	L_n
Trigonal	$3m'$	32	$3m$	L_m	M_m	M_n	L_n	M_m	L_m	L_n	M_n
Trigonal	$\overline{3}m$	—	—	L_m	—	—	L_n	L_m	—	—	L_n
Trigonal	$\overline{3}m'$	$\overline{3}m$	$3m$	L_m	—	—	L_n	M_m	—	—	M_n
Trigonal	$\overline{3}'m'$	32	$\overline{3}m$	L_m	—	—	L_n	—	L_m	L_n	—
Trigonal	$\overline{3}'m$	$3m$	$\overline{3}m$	L_m	—	—	L_n	—	M_m	M_n	—

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System	Magnetic point group	Assoc. group A	Assoc. group B	**i-tensor** Polar even m	Axial even m	Polar odd n	Axial odd n	**c-tensor** Polar even m	Axial even m	Polar odd n	Axial odd n
Hexagonal	6	—	—	N_m	N_m	N_n	N_n	N_m	N_m	N_n	N_n
Hexagonal	$6'$	$\bar{6}$	$\bar{6}$	N_m	N_m	N_n	N_n	O_m	O_m	O_n	O_n
Hexagonal	$\bar{6}$	—	—	N_m	O_m	O_n	N_n	N_m	O_m	O_n	N_n
Hexagonal	$\bar{6}'$	6	$\bar{6}$	N_m	O_m	O_n	N_n	O_m	N_m	N_n	O_n
Hexagonal	$6/m$	—	—	N_m	—	—	N_n	N_m	—	—	N_n
Hexagonal	$6'/m'$	$6/m$	$\bar{6}$	N_m	—	—	N_n	O_m	—	—	O_n
Hexagonal	$6/m'$	6	$6/m$	N_m	—	—	N_n	—	N_m	N_n	—
Hexagonal	$6'/m$	$\bar{6}$	$6/m$	N_m	—	—	N_n	—	O_m	O_n	—
Hexagonal	622	—	—	P_m	P_m	P_n	P_n	P_m	P_m	P_n	P_n
Hexagonal	$6'22'$	$(\bar{6}2m)$	$(\bar{6}2m)$	P_m	P_m	P_n	P_n	(R_m)	(R_m)	(R_n)	(R_n)
Hexagonal	$62'2'$	$6mm$	$6mm$	P_m	P_m	P_n	P_n	Q_m	Q_m	Q_n	Q_n
Hexagonal	$6mm$	—	—	P_m	Q_m	Q_n	P_n	P_m	Q_m	Q_n	P_n
Hexagonal	$6'mm'$	$\bar{6}m2$	$(\bar{6}2m)$	P_m	Q_m	Q_n	P_n	(R_m)	R_m	R_n	(R_n)
Hexagonal	$6m'm'$	622	$6mm$	P_m	Q_m	Q_n	P_n	Q_m	P_m	P_n	Q_n
Hexagonal	$\bar{6}m2$	—	—	P_m	R_m	R_n	P_n	P_m	R_m	R_n	P_n
Hexagonal	$(\bar{6}'2m')$	622	$(\bar{6}2m)$	P_m	R_m	R_n	P_n	(R_m)	P_m	P_n	(R_n)
Hexagonal	$\bar{6}'m2'$	$6mm$	$(\bar{6}2m)$	P_m	R_m	R_n	P_n	(R_m)	Q_m	Q_n	(R_n)
Hexagonal	$\bar{6}m'2'$	$\bar{6}m2$	$6mm$	P_m	R_m	R_n	P_n	Q_m	R_m	R_n	Q_n
Hexagonal	$6/mmm$	—	—	P_m	—	—	P_n	P_m	—	—	P_n
Hexagonal	$6'/m'mm'$	$6/mmm$	$(\bar{6}2m)$	P_m	—	—	P_n	(R_m)	—	—	(R_n)
Hexagonal	$6/mm'm'$	$6/mmm$	$6mm$	P_m	—	—	P_n	Q_m	—	—	Q_n
Hexagonal	$6/m'm'm'$	622	$6/mmm$	P_m	—	—	P_n	—	P_m	P_n	—
Hexagonal	$6/m'mm$	$6mm$	$6/mmm$	P_m	—	—	P_n	—	Q_m	Q_n	—
Hexagonal	$6'/mm'm$	$(\bar{6}2m)$	$6/mmm$	P_m	—	—	P_n	—	(R_m)	(R_n)	—
Cubic	23	—	—	S_m	S_m	S_n	S_n	S_m	S_m	S_n	S_n
Cubic	$m\bar{3}$	—	—	S_m	—	—	S_n	S_m	—	—	S_n
Cubic	$m'\bar{3}$	23	$m\bar{3}$	S_m	—	—	S_n	—	S_m	S_n	—
Cubic	432	—	—	T_m	T_m	T_n	T_n	T_m	T_m	T_n	T_n
Cubic	$4'32'$	$\bar{4}3m$	$\bar{4}3m$	T_m	T_m	T_n	T_n	U_m	U_m	U_n	U_n
Cubic	$\bar{4}3m$	—	—	T_m	U_m	U_n	T_n	T_m	U_m	U_n	T_n
Cubic	$\bar{4}'3m'$	432	$\bar{4}3m$	T_m	U_m	U_n	T_n	U_m	T_m	T_n	U_n
Cubic	$m\bar{3}m$	—	—	T_m	—	—	T_n	T_m	—	—	T_n
Cubic	$m\bar{3}m'$	$m\bar{3}m$	$\bar{4}3m$	T_m	—	—	T_n	U_m	—	—	U_n
Cubic	$m'\bar{3}m'$	432	$m\bar{3}m$	T_m	—	—	T_n	—	T_m	T_n	—
Cubic	$m'\bar{3}m$	$\bar{4}3m$	$m\bar{3}m$	T_m	—	—	T_n	—	U_m	U_n	—